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The Missing Dimension of Computing

*Why Computers Understand Data but Not Value***Hassler Juma & Joseph Njoroge**

Co-Founders, Senti AI

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Abstract

For nearly seventy years, computer science has gotten remarkably good at one thing: moving data around. We have built operating systems that manage hardware, databases that structure records, networks that carry packets, and now artificial intelligence that finds patterns in almost anything. And yet, for all of that scaffolding, a modern computer still has no idea what value is. To a machine, a transaction is just a number changing in a database. An environmental cost is a column nobody reads. A human life-and-death priority is a variable nobody thought to define.

This entry argues that the split between how computers process information and how they understand value is the deepest blind spot in modern computing, and that it is time to name it, study it, and build for it. We introduce Value Engineering as a new, foundational branch of computer science: the study of how machines can represent, preserve, transform, and reason about value. We place it in the natural line of computing's evolution, from numbers, to records, to documents, to language, and now to value itself, and close by introducing Daturrency, a first working theory for turning passive digital money into something that actually understands what it is for.

Keywords: value engineering · computational value · Daturrency · scalar representation bottleneck · DIKW hierarchy · value-aligned AI · knowledge objects · digital public finance

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Editor's note on Entry 000

This is the founding entry of the Value Engineering Journal, a series built around one question computer science has mostly left unexamined: what would it take for a machine to actually understand value, rather than just move data around?

Entry 000 lays the theoretical ground the rest of the series will build on. It names Value Engineering as a discipline and introduces Daturrency as its first working theory. Later entries will take these ideas further, into implementation, evaluation, and field application.

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01 The Illusion of Digital Wealth

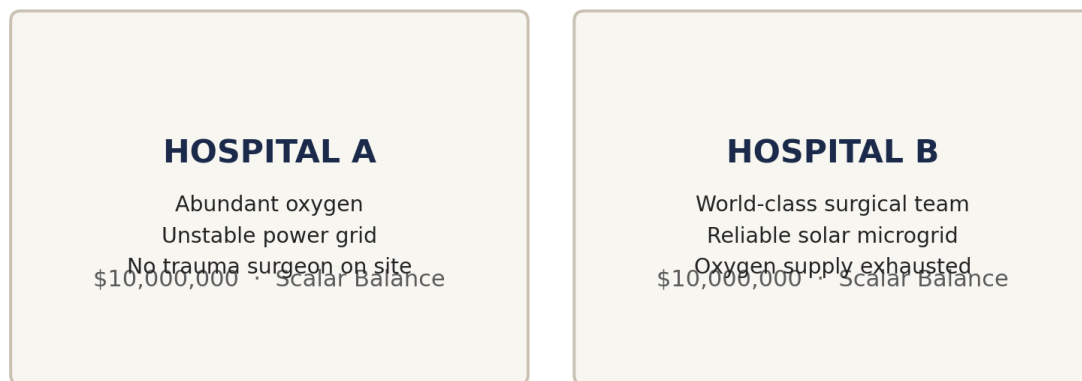
Take two regional hospitals, each holding exactly ten million shillings on their digital bank ledgers.

To the global financial system, its relational databases, its core banking APIs, these two institutions look identical. Same wealth, same creditworthiness, same economic standing. On paper, they are equals.

Step outside the ledger, though, and the picture changes completely. Hospital A sits in a region with plenty of liquid oxygen, but its power grid is unreliable and it has no trauma surgeon on site. Hospital B has a rock-solid private solar microgrid and a world-class surgical team, but its oxygen supply just ran dry because of a regional supply chain disruption.

Under the ledger systems we currently run on, these two hospitals cannot simply find each other and trade. Their bank balances are silent. A computer looking at those accounts has no way to see the missing power at Hospital A, the missing oxygen at Hospital B, or the simple fact that a surgeon's time cannot be swapped for anything else.

Because the system underneath only understands one thing, the balance, a plain number, it cannot see that trading Hospital B's surplus oxygen for Hospital A's surgical access would save lives on both sides, almost instantly. Instead, both hospitals are pushed into slow, separate, cash-mediated procurement queues. Patients die at Hospital A for want of surgery. Patients die at Hospital B for want of oxygen. And both ledgers keep displaying a calm, healthy balance of ten million.



Identical on the ledger. Not identical in reality.

Figure 1. Identical ledger balances, opposite physical realities.

This is not a failure of hospital administration. It is a failure of our foundational computational abstractions. Humans read situations like this instantly, we weigh context, anticipate consequences, and balance competing priorities almost without trying. Computers cannot. To a computer, data has no depth. It is a sequence of bytes moving from an input register to an output register, and nothing tells it what is worth preserving, what is urgent, or what happens in the physical world because of the move it just made.

The world runs on value. Computers only understand data.

02 Money Is Not Value: The Scalar Representation Bottleneck

Before we go further, we need to clear up a confusion that sits underneath almost everything else here: money is not value.

Money is a compressed, lossy shorthand for value, a tool built to make exchange possible in a world with limited bandwidth for complexity. Every time we flatten the many-sided reality of human health, environmental stability, resource availability, and time into a single number, we are making a trade-off. We call this the Scalar Representation Bottleneck.

Picture four things of equal market price sitting in the same local economy: one hundred dollars in cash, one litre of clean drinking water, one hour of a doctor's diagnostic time, and one kilogram of organic crop seed.



Four systems. Zero shared context.

Figure 2. Four resources of equal price, stored in four systems that never talk to each other.

To any modern computer system, these four things live in completely separate worlds. The cash is a row in a bank's SQL table. The water is a reading on a municipal IoT dashboard. The doctor's hour is a block on a hospital calendar. The seed is a SKU in a supply chain database. None of these systems talk to each other, so no computer anywhere can weigh their relative worth in context.

To a community facing famine, that kilogram of seed is future survival, worth infinitely more than the cash sitting next to it. To an overcrowded clinic mid-outbreak, the doctor's hour is the one thing that cannot be substituted for anything else. To someone dying of thirst in an arid region, the water is the only thing standing between them and death, and the cash is just paper.

Value is a high-dimensional, context-dependent space. It includes things like the health of an aquifer, the carbon locked in topsoil, and the biodiversity of a region, which we might call ecology. It includes the non-recoverable hours of human attention and labour, which is time. It includes the strength of social contracts and community trust, which is trust. And it includes the nutritional and healthcare baseline a population can count on, which is health.

Money is just one coordinate inside that much larger space. When we build global systems that only track the monetary coordinate, we build engines of lossy reductionism, systems that maximise transaction speed and ledger throughput while quietly grinding down the physical networks that keep us alive.

03 The Three Great Abstractions

To place a formal science of value inside the broader map of human knowledge, it helps to think of the universe in three layers.

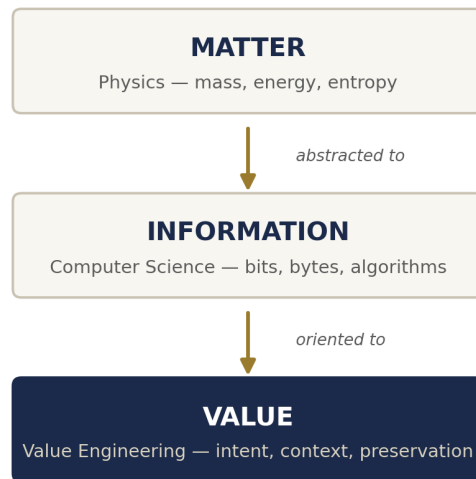


Figure 3. From Matter, to Information, to Value: the next abstraction layer in computing.

Matter is the physical universe, governed by thermodynamics, physics, and chemistry. It is about mass, energy, entropy.

Information is the symbolic layer sitting on top of matter, governed by computer science, information theory, and computation. It is about bits, bytes, complexity, and how algorithmic states move around.

Value is the purposeful layer, governed by what we are calling Value Engineering. It is about human intent, contextual worth, resource preservation, and the outcomes a system actually produces.

Computer science came into being because physics and electrical engineering, on their own, could not fully explain what software does. In the same way, Value Engineering needs to exist because computer science, on its own, cannot fully explain what value does. We have spent seventy years learning to move from Matter to Information. The next seventy start with moving from Information to Value.

04 The Arc of Computational Representation

This blind spot is not a design flaw so much as a historical accident. Every era of computing has been shaped by whatever bottleneck it was fighting at the time. Looking back, there is a clear, almost inevitable widening of what machines are able to work with.



Figure 4. Each era of computing solved for a different bottleneck.

The Number Era of the 1950s had hardware as its bottleneck. Computers were fast calculators built for scalar arithmetic and military trajectory math. Money was represented as a plain integer because anything richer would have blown the memory budget.

The Record Era of the 1970s had storage as its bottleneck. Relational algebra gave us databases that could store and query structured rows and columns.

The Document Era of the 1990s had connectivity as its bottleneck. The web arrived to link scattered files and documents into one searchable, human-readable network.

The Language Era of the 2010s had context as its bottleneck. Machine learning and large language models let computers move from matching raw strings to statistically understanding language.

The Value Era, which we would place in the 2030s, has purpose and consequence as its bottleneck. Compute, storage, and reasoning power are abundant now. What is missing is alignment: how do we teach machines to understand, preserve, and optimise for what actually matters to human and ecological survival?

Treating value as a plain scalar number was a reasonable compromise in 1960. In 2026, it is an obsolete one.

05 Data Is Not Value: The Value-Knowledge Hierarchy

Information systems talk a lot about the DIKW pyramid: Data, Information, Knowledge, Wisdom. Below is a version of that hierarchy built specifically for computational value systems, tracing the path from raw signal to something worth protecting.

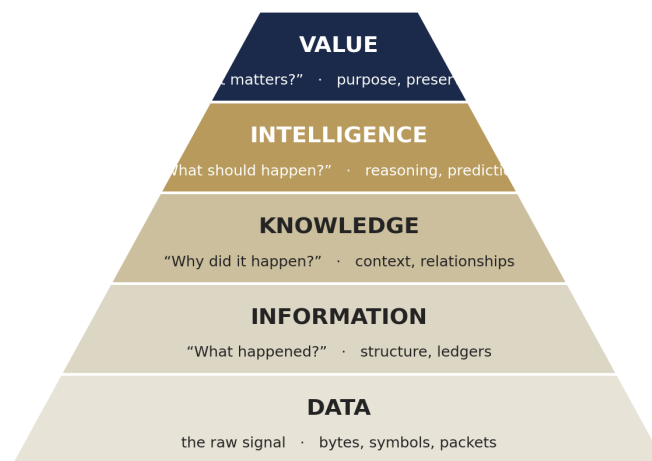


Figure 5. The Value-Knowledge Hierarchy, extending DIKW with a teleological cap.

Data is the raw signal. It is just bits: 01001011.

Information gives data shape. It turns bits into a ledger line: “John transferred 100 to Mary.”

Knowledge connects information to context. It explains why the transfer happened and what depends on it: “John paid Mary for fertiliser because his soil is short on nitrogen before planting season.”

Intelligence uses that knowledge to reason forward: “If John skips the fertiliser today, his yield drops forty percent, and his household is short on food in six months.”

Value sits at the top. It defines what actually matters, the compass, the goals, and the boundaries that should guide the intelligence layer: “Protect John's household nutrition. Protect the soil from chemical runoff. Do not deplete the water table.”

Today, our systems are excellent at the Data, Information, and Knowledge layers, and, thanks to modern AI, closing in fast on the Intelligence layer. But the Value layer is still almost entirely missing. We have built remarkably smart systems and given them no native, machine-readable sense of what is actually worth protecting. We have built fast cars with no steering wheel, driving on a map that shows coordinates but no destination.

06 Why Artificial Intelligence Alone Cannot Solve the Value Problem

A natural objection here is: will AI not just solve this on its own? If we have capable large language models and autonomous agents, can they not simply handle these trade-offs for us?

No, and it is worth being direct about why.

AI models are excellent at reasoning over language, spotting patterns, and executing API calls. But they are not value-native. Ask an AI agent to buy fertiliser for a farmer, and it will search the web, find the cheapest price, and fire off the transaction through a bank API.

Task complete, but the agent stays completely blind to what happens next. It does not know the local water table is already depleted and this specific fertiliser will leach into the community's main well. It does not know the farmer's child has a medical bill due next week, and spending on fertiliser today creates a healthcare crisis tomorrow. It does not know a drought is forecast in three months, which would make drought-resistant seed a far better use of that same money.

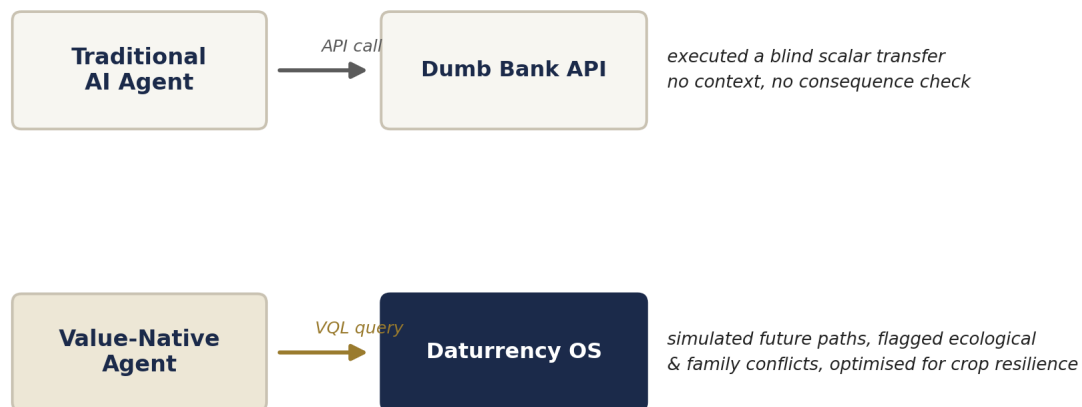


Figure 6. A traditional agent executes; a value-native agent reasons over consequences.

An AI agent is only ever as good as the data model underneath it. If that model represents money as a blind, contextless number, the AI has no choice but to act blindly too.

An AI reasons over language. A value-computing system reasons over consequences. Building safe, human-aligned, ecologically sane systems is not just a matter of making agents smarter, it requires giving them an intelligent data model for value in the first place. The AI needs to be able to read, query, and reason over value as a structured, goal-aware object, not guess at it from a bank balance.

07 The Birth of a New Discipline: Value Engineering

We are proposing Value Engineering as a formal, interdisciplinary branch of computer science.

The same way Database Systems emerged to study how we represent and query structured records, and Machine Learning emerged to study how machines learn from data, Value Engineering exists to study how computers represent, preserve, transform, and reason about value in an intelligent world.

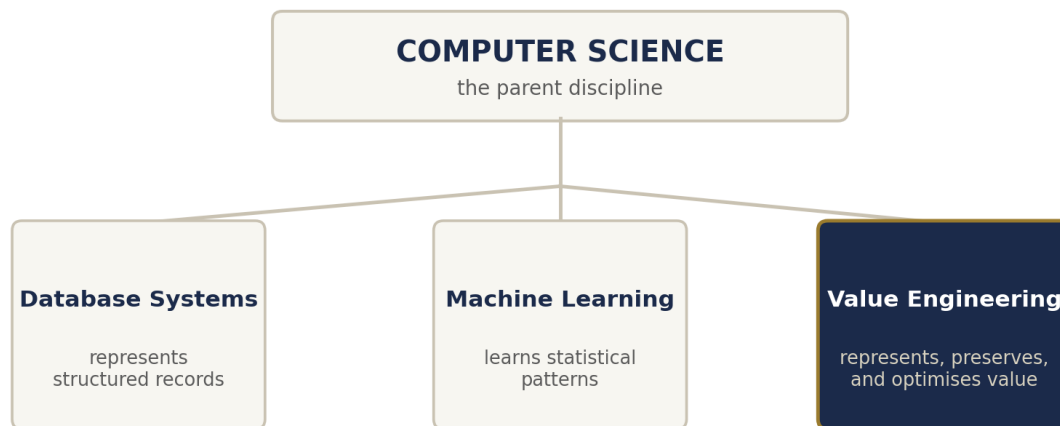


Figure 7. Value Engineering positioned alongside Database Systems and Machine Learning.

It sits at the intersection of several existing fields. Systems Ecology and Cybernetics contribute feedback loops, resource flows, and self-regulating systems. Information Systems and the Semantic Web contribute rich ontologies, metadata graphs, and machine-readable context. Multi-Agent Systems and Decision Theory contribute algorithms for autonomous planning and multi-objective optimisation. Monetary Economics contributes a rethink of the computational mechanics of transactions, wealth tracking, and resource allocation.

Value Engineering does not replace any of these fields. It unifies them, giving their combined insight a computational engine that can actually run in real time, at the kernel level of our global systems.

08 Introducing Daturrency: Money as an Operating System Object

Within Value Engineering, we present Daturrency as the foundational theory.

Daturrency is not a new currency. It is not a blockchain token, not a Central Bank Digital Currency, and not a private stablecoin. It is an abstraction layer, a computational framework that sits on top of the financial databases, mobile money networks, and blockchains we already have, and turns passive digital cash into an active, goal-aware knowledge object.

Under Daturrency, a unit of money stops being a blind number on a ledger. It starts behaving more like an operating system object, something closer to a process or a file, carrying its own permissions and metadata.

DATURRENCY KNOWLEDGE OBJECT	
Magnitude:	KES 50,000
Provenance:	Local Agroforestry Cooperative
Intent:	Land Restoration and Household Nutrition
Constraints:	Geofenced to Rift Valley · No Alcohol · Monotonic decay after 90 days
Trajectory:	Compound organic soil health accumulation

Figure 8. A Daturrency object: money carrying its own metadata, intent, and constraints.

By packaging money together with metadata, intent, and constraints, the money itself becomes an active participant in a person's financial life, one that owes them a kind of fiduciary duty.

When you get paid, instead of your salary sitting there as an inert pool waiting to be spent on impulse, it is self-allocating. It knows you are saving for a home. It knows your medical priorities. It knows the environmental constraints of your local community. It negotiates bills, schedules payments, chooses food for nutritional value, and protects your long-term resilience, all while staying under your continuous, sovereign control.

09 Why This Matters: The Real-World Impact

Once we stop treating value as an afterthought and start treating it as data, some genuinely old problems start to look solvable.

9.1 Breaking the Cognitive Scarcity Loop

We know from psychology and decision theory that financial scarcity eats up enormous amounts of cognitive bandwidth. A low-income worker runs the same exhausting calculation every day: pay the electricity bill today, or buy better food for the kids? Can I even afford to put aside a bit for school fees?

That constant mental math drains the cognitive equivalent of working memory, and keeps people trapped in survival mode.

Daturrency moves that burden off the human brain and onto the computational substrate. Running quietly on a person's phone, it calculates those trade-offs itself, reserving cash for critical bills, finding cheaper agricultural inputs, and routing small amounts toward long-term goals, without the person having to relive that decision fatigue every single day.

9.2 Complete Public Finance Accountability

Right now, taxation is close to a black box. Citizens pay into a central treasury, and from that point on, the money loses its history, its intent, its trail. Governments can spend heavily repaving roads that do not need it while regional clinics sit with empty shelves.

If public finance ran on Daturrency, tax contributions would keep their Intent Preservation property. Money earmarked by a citizen for healthcare would move through the public value graph as a constrained, traceable object, one that literally cannot be redirected into an unrelated road

contract or administrative overhead, because the database itself blocks any transfer that violates the original intent.

9.3 Ecological-Economic Harmony

By feeding ecological sensor data directly into the value graph, we get economies that can regulate themselves. As a regional aquifer runs low, the local value engine automatically raises the virtual cost of water-heavy industrial permits and starts subsidising drought-resistant seed swaps, without anyone needing to intervene manually.

Economy and ecology, finally, sit inside the same self-balancing loop.

10 Conclusion: The Next Era of Computing

The story of computing has always been a story of expanding what we can represent.

We taught computers to calculate. Then we taught them to communicate. Then we taught them to learn.

The next step is not making them faster, smaller, or more powerful. It is giving them a language for what actually matters.

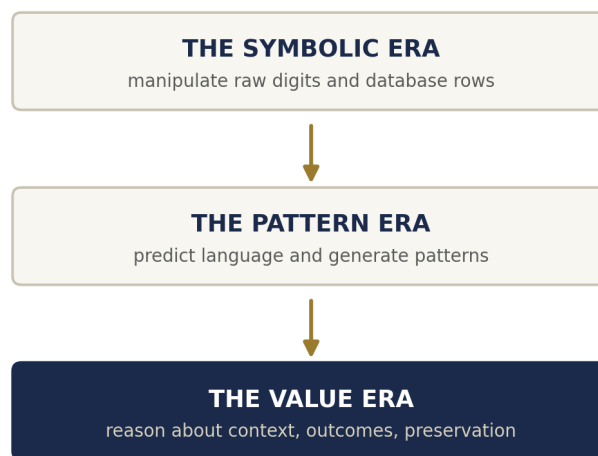


Figure 9. Three eras of computational representation, from symbols to values.

Computers reshaped human society once already, by learning to process data and information. They may reshape it again, this time by learning to reason about what is worth preserving.

Value Engineering, and the theory of Daturrency introduced here, are meant as a starting architecture for that shift. If we can move past the reductionism of scalar, double-entry ledgers, we can build a version of technology that does not just move faster, it preserves, optimises, and honours the complex, living networks we all depend on.

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